

ASSIGNMENT REPORT – 2

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For the Subject

PATTERN RECOGNITION AND MACHINE LEARNING
(26AI5707)



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Introduction

- In this lab experiment, we studied Eigenvalue Decomposition (EVD) and Singular Value Decomposition (SVD) and applied them to reconstruct the assigned images. First, the images were converted into grayscale and then represented as numerical matrices, where each element stands for the intensity of a pixel.
- SVD is commonly used for dimensionality reduction, image compression, noise reduction, and low-rank approximation. EVD is commonly used in applications such as Principal Component Analysis (PCA), matrix powers, differential equations, and signal processing.
- Two images were used in this experiment:
 - Square Image (512×512): Both EVD and SVD were performed on the square image. The images were reconstructed using selected components, and the original, reconstructed, and error images were displayed. The reconstruction error was calculated using the Frobenius norm.
 - Rectangular Image (337×600): SVD was performed on the rectangular image. SVD can be applied to rectangular matrices, while standard Eigenvalue Decomposition needs a square matrix.

Experiments Performed

Task 1: Eigenvalue Decomposition (EVD) and Singular Value Decomposition (SVD) of the Square Image

The given square image was first converted into grayscale. The grayscale image was then represented as a numerical matrix:

$$A \in \mathbb{R}^{(512 \times 512)}$$

Each element of the matrix stands for the intensity of a pixel in the grayscale image.

Eigenvalue Decomposition (EVD)

For square matrix A, Eigenvalue Decomposition is represented as:

$$A = Q\Lambda Q^{-1}$$

Where Q is the matrix of eigenvectors and Λ is a diagonal matrix having the eigenvalues. The eigenvalues and eigenvectors of the image matrix were calculated using the Eigenvalue Decomposition method. The eigenvalues were then sorted according to their absolute magnitude. The code also checks for complex conjugate eigenvalue pairs. If a complex eigenvalue is encountered, its corresponding conjugate eigenvalue is found and considered along with it during the ordering process. For a selected value of k, a new diagonal matrix Λ_k was created by keeping the first k eigenvalues and setting the remaining eigenvalues to zero. The reconstructed matrix was then calculated as:

$$A_k = Q\Lambda_k Q^{-1}$$

Since the square image matrix is not necessarily symmetric, complex eigenvalues and eigenvectors may occur. Therefore, after reconstruction, the real part of the reconstructed matrix was considered. The reconstructed pixel values were also restricted to the grayscale range of 0 to 255.

Singular Value Decomposition (SVD)

The Singular Value Decomposition of matrix A is represented as:

$$A = U \Sigma V^T$$

where U is the left singular-vector matrix, Σ is a diagonal matrix having the singular values, and V^T is the transpose of the right singular-vector matrix. The singular values obtained from SVD are arranged in decreasing order. Therefore, the first few singular values hold the most significant information about the image.

For a selected value of k , the reconstructed matrix is given by:

$$A_k = U_k \Sigma_k V_k^T$$

where only the first k singular values and their corresponding vectors are kept. The square image was reconstructed using the following values of k , considering values with the least Frobenius error and with a compression factor (CF) above 1. For each value of k , the original image, reconstructed image, and error image were displayed. The values that satisfy the conditions are:

$$k = 180$$

$$k = 220$$

$$k = 240$$

Compression Factor:

The compression factor (CF) is a metric which is used in this implementation to calculate the ratio between the compressed size of an image with respect to its original size. The Compression Factor is given by:

$$CF = \frac{\text{Compressed Size}}{\text{Original Size}}$$

For $CF > 1$, Image has been enlarged w.r.t. the original image

$CF = 1$, Data has equivalent size w.r.t. the original image

$CF < 1$, Data has been compressed w.r.t. the original image

For an image compressed through EVD, the CF is calculated by:

$$CF_{EVD} = \frac{2k \cdot (n+1)}{n^2}$$

Reconstruction Error

The difference between the original image and the reconstructed image was calculated using the Frobenius norm.

The reconstruction error is:

$$E(k) = \|A - A_k\|_F$$

The Frobenius norm is given by.

$$E(k) = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij} - (A_k)_{ij}|^2}$$

where:

- A is the original image matrix.
- A_k is the reconstructed image matrix.
- M is the number of rows.
- N is the number of columns.

A lower Frobenius error shows that the reconstructed image is closer to the original image.

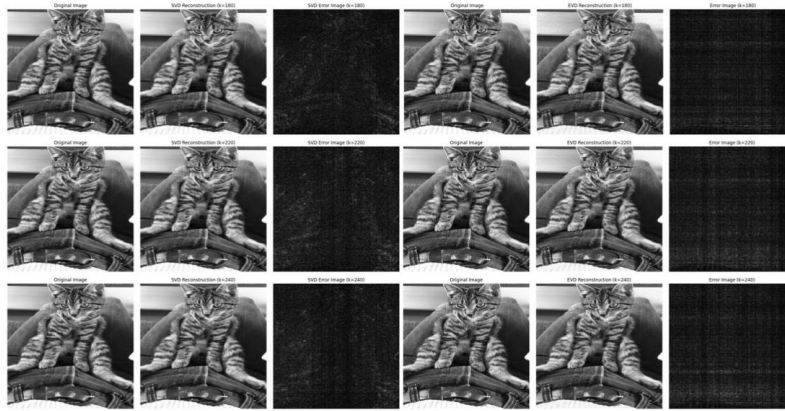


Figure 1: Original image, EVD reconstruction, EVD error map, SVD reconstruction, and SVD error map, for $k = 180, 220, 240$

Compression Calculation for the Squared Image

The original square image is 512×512 so it has 262,144 elements. In SVD, a rank- k approximation stores U_k , Σ_k , and V_k^T instead of the full matrix. The compressed storage is calculated as $2n(k+1)$, where n is the matrix size,

and k is the number of kept singular values. A smaller k needs less storage but loses more image information, showing the trade-off between compression and reconstruction quality.

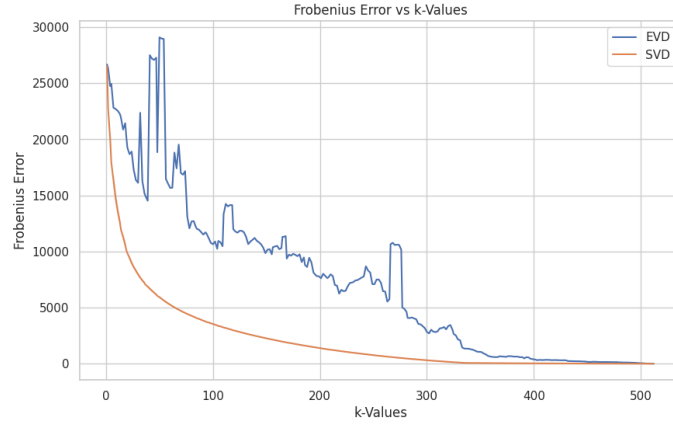


Figure 2: Reconstruction error $E(k)$ versus the number of kept components k , for EVD and SVD.

Task 2: SVD of the Rectangular Image

The second image used in the experiment was the rectangular image. The original image had dimensions of 337×600 . After conversion to grayscale, the image was represented as $A \in \mathbb{R}^{(337 \times 600)}$. Since the matrix is rectangular, SVD was used for its decomposition. The SVD of the rectangular image is given by:

$$A = U\Sigma V^T$$

The maximum possible rank of the matrix is $\min(337, 600) = 337$. The rectangular image was reconstructed using the values of $k = [180, 190, 200, 210, 220, 230, 240, 250]$. For every selected value of k , the first k singular values were kept, and the corresponding reconstructed image was generated. The reconstructed matrix was then restricted to the valid grayscale range of 0 to 255. For every value of k , the original image, reconstructed image, and error image were displayed.

Compression Factor

The original rectangular image is 337×600 , so it has 202, 200 elements. For a rank- k SVD representation, the storage requirement is calculated as $mk + k + kn$, where $m = 337$, $n = 600$, and k is the number of kept singular values. Instead of storing the complete image matrix, SVD stores the required parts of U , Σ , and V^T , allowing the image represented using fewer components. The Compression Factor for Singular Value Decomposition (SVD) is given by:

$$CF_{SVD} = \frac{m \cdot n}{k \cdot (m+n+1)}$$

Reconstruction error for the Rectangular Image

The reconstruction error for the rectangular image was calculated using the Frobenius norm $E(k) = \|A - A_k\|_F$. The error image was calculated as $|A - A_k|$

The error values were stored in a table along with the corresponding values of k . As the number of kept singular values changes, the amount of information kept from the original image also changes. This allows the reconstruction quality to be compared for different values of k .

Comparison of EVD and SVD

TABLE 1: Comparison of EVD and SVD

Feature	EVD	SVD
Decomposition	$A = Q\Lambda Q^{-1}$	$A = U\Sigma V^T$
Matrix type	Square matrices	Square and rectangular matrices
Values	Eigenvalues can be complex	Singular values are real and non-negative
Low-rank approximation	Less suitable	Highly suitable
Image compression	Less commonly used	Commonly used
Image reconstruction	Square matrices	Square and rectangular matrices

INFERENCE

The experiment showed how EVD and SVD can be used for image reconstruction. The images were first converted into grayscale matrices and then decomposed into their respective components. The reconstruction quality was measured using the Frobenius error. SVD was especially useful because it can be applied to both square and rectangular matrices. By changing the value of k , we can control the balance between image quality and the amount of storage needed.

CONCLUSION

This experiment proved the use of EVD and SVD for image reconstruction and compression. While both EVD and SVD were used in square image sample, only SVD was used in rectangular image sample as EVD requires square image. After reconstruction of two images, with both the decomposition methods, SVD tends to reconstruct with lesser Frobenius error for the same amount of k values compared to EVD.